

Data Collection and Preprocessing Phase

Date	22 July 2026
Team ID	Priya Kumari
Project Title	Global AI adoption in Education
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains global AI adoption statistics in education across multiple countries and months. It includes information on student AI usage, teacher AI usage, school AI adoption, average daily AI usage, top AI tools, internet penetration, education index, government AI policies, urban and rural AI usage, AI curriculum integration, and gender gap in AI usage.
Data Cleaning	Missing Values: Identified missing values in columns such as Government AI Policy and replaced them with appropriate values (e.g., "Unknown") or removed incomplete records where necessary. Duplicates: Checked and removed duplicate records to maintain data integrity. Error Correction: Corrected inconsistent spellings, removed extra spaces, standardized country and region names, and verified percentage values were within valid ranges (0–100).
Data Transformation	Calculated Fields: Created new calculated metrics such as Total AI Usage, Urban-Rural Usage Difference, and Student-Teacher Usage Ratio. Aggregations & Pivot Tables: Used Pivot Tables and summary statistics to analyze AI adoption by country, region, and year. Filtering: Removed invalid or extreme values that could negatively affect the analysis and retained only meaningful observations.
Data Type Conversion	Converted Year and Month columns to integer format, percentage-related columns to numeric data types, Education Index to decimal format, Government AI Policy and AI in Curriculum to categorical variables, and

	Country, Region, and Top AI Tool to text data types for accurate visualization and analysis.
Column Splitting and Merging	Standardized country and region information where necessary. Combined Year and Month columns into a Date field for time-series analysis. Created additional derived columns such as AI Adoption Category and Internet Accessibility Level for improved dashboard visualization.
Data Modeling	Organized the cleaned dataset into a structured format suitable for Tableau/Power BI. Established relationships between geographical, educational, and AI adoption attributes to support comparative analysis across countries and regions. Calculated summary measures such as average AI usage, regional adoption rates, and curriculum implementation percentages.
Save Processed Data	Saved the cleaned and transformed dataset as a CSV/Excel file with standardized column names and formats. Verified the dataset before importing it into Tableau/Power BI and ensured all columns were correctly recognized for visualization.